

Prime-Indexed Multiplicity Cobordism Invariants (PIMCI)

Canonicalization (“Kirby gauge”), smooth sensitivity, and a minimal validation pipeline

Context. This document distills a checkable mathematical core for your **prime-indexed cobordism invariants** idea in the Multiplicity / MOC style:

- primes label operator families and “tiers” are extracted by idempotent projectors Π_{p^r} ,
- cobordisms/handle moves act as noncommuting operator words,
- adequacy can be evaluated by a resonance functional $R \in [0, 1]$.

Where PIMCI becomes genuinely *mathematical* (and not presentation-dependent) is the **canonicalization step**: invariants are computed only after reducing the chosen surgery/handle presentation to a **canonical representative** of its Kirby/handle-equivalence class.

1. Core problem: handle slides act by *congruence*, not similarity

For a framed-link / 2-handle presentation producing an integral symmetric matrix $K \in M_m(\mathbb{Z})$ (linking/intersection data), a handle slide corresponds algebraically to

$$K \mapsto U^{-1} K U, \quad U \in GL(m, \mathbb{Z}),$$

where U is an elementary matrix.

Therefore, raw “spectral” quantities like $\text{Tr}(K^k)$ are *not* invariants unless we first pass to a **canonical representative** of the stable congruence class.

2. Kirby gauge: explicit generating moves and canonicalization target

2.1 Gauge group generators (moves)

Start from a surgery/plumbing presentation $\mathcal{P}$ of a (homotopy) sphere Σ^n that yields an integral symmetric matrix K .

Let the **gauge group** be generated by:

1. Permutation / relabeling

$$K \mapsto P K P^{-1}, \quad P \text{ a permutation matrix.}$$

1. **Orientation flip** (reverse orientation of a component/handle)

$$K \mapsto D K D, \quad D = \mathrm{diag}(\pm 1, \dots, \pm 1).$$

1. **Handle slide** $i \leftarrow i + \varepsilon j$ with $\varepsilon \in \{\pm 1\}$

$$K \mapsto U_{\{ij\}}(\varepsilon) K U_{\{ij\}}(\varepsilon)^{-1}, \quad U_{\{ij\}}(\varepsilon) = I + \varepsilon E_{\{ij\}}.$$

1. **Stabilization** (Kirby I blow-up/down)

$$K \mapsto K \oplus \pm 1$$

(and the inverse move when a ± 1 summand splits off).

These generate the standard equivalence of framed-link presentations relevant for computing stable invariants.

2.2 Canonicalization target

Define the canonicalization target as the **stable isometry class** of the quadratic lattice

$$(\mathbb{Z}^m, \text{form}) \quad \text{(+ optional refinements: spin/characteristic data, smooth defect, etc.)}$$

Two presentations are equivalent iff their lattices are related by the above moves (congruence + stabilization + relabeling/orientation).

3. Practical canonicalization: do it prime-by-prime

A robust, computable canonical “name” for K can be produced using local classification:

Step A — normalize parity/spin constraints

If working in the spin/framed setting, stabilize (via $\oplus[\pm 1]$) until the parity condition (even/odd, characteristic element constraints) matches the intended bordism structure.

For each prime p , classify the p -adic completion up to $GL(m, \mathbb{Z}_p)$ -congruence:

$$K \otimes \mathbb{Z}_p \cong \bigoplus_{r \geq 0} p^r K_{\{p, r\}},$$

with $K_{p, r}$ unimodular over $\mathbb{Z}_p$.

Record the canonical **symbol**

$\mathsf{NF}_p(K) = \{(r, \mathrm{rank}(K))_r\}$, \text{unit/Hasse-type data}

This gives a **prime-indexed**, move-invariant descriptor of the stable congruence class.

Step C — the real place

Record $\sigma(K \otimes \mathbb{R})$ (signature) and parity; these constrain global reconstruction.

Step D — optional reconstruction of a canonical integral matrix

Optionally, reconstruct a canonical integral representative K_{can} from $\{\mathsf{NF}_p(K)\}_p$ and σ , using a deterministic “lexicographically minimal” construction.

Output of canonicalization (“Kirby gauge”):

$\mathrm{Can}(\mathcal{P}) = \Big(\{\mathsf{NF}_p(K)\}_p, \sigma, \text{parity/refinements}\Big)$.

Why you need this

Many invariants extracted from K alone are homotopy-level. To detect **exotic smooth structure** on spheres, you must include a smooth-sensitive term.

Dimension 7: the Eells–Kuiper invariant

For homotopy 7-spheres, the renormalized Eells–Kuiper invariant $\mu(\Sigma) \in \mathbb{Z}/28$ classifies oriented diffeomorphism types (equivalently, it detects the 28 exotic smooth structures).

A minimal integration into PIMCI is to attach a scalar defect $\delta(\Sigma) = \mu(\Sigma)$ and augment the multiplicity matrix:

$\widetilde{M}_\Sigma = \begin{pmatrix} M_\Sigma & 0 \\ 0 & \delta(\Sigma) \end{pmatrix}$.

Prime-primary structure becomes explicit because

$\mathbb{Z}/28 \cong \mathbb{Z}/4 \times \mathbb{Z}/7$,

so only primes 2 and 7 carry the classification signal at the smooth level.

Define Π_{p^r} on canonical symbols (not on a basis-dependent matrix):

$\Pi_{p^r}(\mathsf{NF}_p(K)) := \{(r, \mathrm{rank}(K), \dots)\}$.

This is:

- **idempotent** by definition,
- **compatible with gauge** because it acts after quotienting,
- inherently **prime-indexed**.

If you reconstruct K_{can} , you may also apply Π_{p^r} to K_{can} and then compute traces/determinants safely.

6. PIMCI invariants (canonical-symbol first; matrix-spectral optional)

Canonical-symbol invariants (recommended)

For each prime power p^r , define invariants directly from the canonical symbol and smooth defect:

- block ranks $\text{rank}(K_{p,r})$,
- unit/Hasse-type data,
- parity/characteristic data,
- smooth defect components $\mu \bmod 4$ and $\mu \bmod 7$ in dimension 7.

Package as a prime-spectrum:

$$\mathfrak{I}(\Sigma) = \big\{ \mathfrak{I}_{p^r}(\Sigma) \big\}_{p,r}.$$

Optional spectral layer (if you insist on traces/determinants)

Once K_{can} (or $\widetilde{M}_{\Sigma, \text{can}}$) is fixed canonically:

$$\mathfrak{I}_{p^r, k}(\Sigma) = \text{Tr} \big((\Pi_{p^r} \widetilde{M}_{\Sigma, \text{can}})^k \big), \\ \mathfrak{J}_{p^r}(\Sigma) = \det(I - s \Pi_{p^r} \widetilde{M}_{\Sigma, \text{can}}).$$

7. Cobordisms as operator words + strict invariance via canonicalization

Encode a cobordism $W : \Sigma_0 \rightsquigarrow \Sigma_1$ as an operator word W built from prime-indexed noncommuting generators.

Instead of trying to force W -independence directly, enforce invariance by definition:

$$\mathfrak{I}(\Sigma) := \mathfrak{I} \big(\text{Can}(\mathcal{P}(\Sigma)) \big).$$

Since Can quotients by Kirby/handle equivalence, $\mathfrak{I}$ is presentation-independent.

8. Resonance functional as a *distance on canonical data*

Turn “resonance” into a mathematically strict notion:

- Define a discrepancy d between canonical datasets,
- set $R = \exp(-d) \in (0, 1]$,
- then $R = 1$ iff canonical data match.

Example sketch:

$$d(\Sigma_0, \Sigma_1) = \sum_p w_p |d_p(\mathrm{NF}_p(K_0), \mathrm{NF}_p(K_1))| + w_\infty |\Sigma_0 - \Sigma_1| + w_\mu \mathbf{1}[\mu_0 \neq \mu_1].$$

9. Minimal viable experiment (MVP) for validation: Milnor 7-spheres

Goal: verify (i) Kirby-move invariance and (ii) exotic sensitivity.

1. **Dataset.** Choose explicit Milnor spheres Σ with known $\mu(\Sigma)$ values.
2. **Pipeline.** For each Σ :
 3. parse plumbing/surgery data $\Rightarrow K$,
 4. compute $\mathrm{Can}(\mathcal{P}) \Rightarrow \{\mathrm{NF}_p(K)\}_p, \sigma$, parity,
 5. attach $\delta(\Sigma) = \mu(\Sigma)$,
 6. compute Π_{2^r}, Π_{7^r} outputs.
7. **Kirby invariance tests.** Randomly apply sequences of algebraic moves $K \mapsto UKU^T$ and $K \mapsto K \oplus [\pm 1]$, then confirm $\mathrm{Can}(\mathcal{P})$ (and therefore $\mathfrak{J}$) is unchanged.
8. **Exotic sensitivity.** Check whether the extracted (2-primary, 7-primary) data correlates exactly with $\mu \bmod 4$ and $\mu \bmod 7$; if yes, PIMCI recovers the correct $\mathbb{Z}/28$ splitting.
9. **Stretch goal.** Move beyond spheres to 2-connected 7-manifolds where generalized invariants refine classification.

10. Open design choices (where “Multiplicity” can add genuine value)

1. **Prime labeling of handles/vertices.** Deterministic labeling (e.g., ordered handles $\mapsto$ first m primes) is fine for MVP, but a principled choice might come from:

2. naturality under refinements,
 3. minimality under canonicalization,
 4. compatibility with operator-word composition.
 5. **Kernel enrichment.** Besides K , incorporate smooth refinements:
 6. characteristic elements / quadratic refinements,
 7. eta-invariant style corrections (where computable),
 8. Pontryagin-class defects in dimensions where they classify.
 9. **Spectral vs symbol invariants.** The symbol-first approach is safest; spectral invariants become meaningful only after a provable canonical matrix representative exists.
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External referrals (for grounding and cross-checking)

Kirby calculus / handle moves

- R. Kirby, *A Calculus for Framed Links in S^3* , 1978 (classic Kirby calculus write-up; operation two = handle slide; intersection form matrix description).
- R. Gompf & A. Stipsicz, *4-Manifolds and Kirby Calculus*, Graduate Studies in Mathematics 20, AMS, 1999. ISBN 9780821809945. DOI: 10.1090/gsm/020.

Exotic spheres and cobordism classification

- M. Kervaire & J. Milnor, *Groups of Homotopy Spheres: I*, *Annals of Mathematics* (2) **77** (1963), 504–537.

Eells–Kuiper invariant and smooth sensitivity (dimension 7)

- J. Eells & N. Kuiper, *An invariant for certain smooth manifolds*, *Annali di Matematica* (1962).
- D. Crowley (and collaborators), works on classification of 2-connected 7-manifolds; includes renormalized Eells–Kuiper invariant $\mu \in \mathbb{Z}/28$. (Example: arXiv:1406.2226).

Matrix representations of handle-slide sequences (optional technical add-ons)

- K. Fujiwara, *Refined Kirby calculus...* (Topology and its Applications; reviews matrix representations of signed handle slides).
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Appendix: what to implement first (engineering checklist)

1. Parser for a plumbing graph / framed link $\Rightarrow K$.
2. Algebraic move engine generating random $U \in GL(m, \mathbb{Z})$ from slides and applying stabilizations.
3. Canonicalization routine outputting $\mathbf{NF}_p(K)$ symbols for small primes (start with $p = 2, 7$).
4. Prime-tier projector Π_{p^r} acting on symbols.
5. Comparison + resonance R computed from canonical data.

(Once this works for Milnor spheres, you can decide whether to promote the operator-word / noncommuting $\hat{O}_p^{(r)}$ layer from “interpretation” to “formal engine.”)

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